VUEESC 2026 + GESSS Nationwide · Program · Thursday 5 November 2026

All times AEDT (Melbourne). Room 1 = VUEESC: 15-min slots (10 talk + 3 questions + 2 changeover). Room 2 = GESSS Nationwide live stream (GESSS Day 1): 10-min slots (8 talk + 2 questions), timed by the national program. Room 1 pauses for GESSS keynotes. ⇄ = both rooms change over at the same moment, so delegates can switch rooms (every 30 min: 2 VUEESC talks = 3 GESSS talks).

GESSS Melbourne hub, in person: the only room running

	ROOM 1 · VUEESC						ROOM 2 · GESSS Nationwide (live stream)				
Time	Slot	Session	Speaker	Institution	Title	⇄	Slot	Speaker	University	Presenting from	Title
9:30	Registration & coffee										
10:30	Welcome to Country, introduction, and a welcome from Head of School Celia McMichael · Room 1, all delegates										
11:00	GESSS KEYNOTE · GESSS opening, then Steve Hill · streamed in Room 1					⇄	11:00–11:10	**Nicko Wyndham**			Opening remarks
11:10							11:10–11:30	**Steve Hill**	Keynote	Canberra, ACT	**KEYNOTE: Geoscientists! Your nation needs YOU: The Government Geoscientist career**
11:30	11:30–11:45	S1 Crustal fluids, resources and carbon storage	**Nathan Roden**	Monash University	Controls on scapolite genesis in salty orogenic belts: Investigating the thermodynamics and chemistry of scapolite to produce ore deposits in orogenic settings	⇄	11:30–11:40	**Alex Molyneux**	ANU	Canberra, ACT	How basic can you get? Constraining secondary precipitation during Ocean Alkalinity Enhancement.
11:40							11:40–11:50	**Zara Smith**	Queensland University of Technology	Brisbane, QLD	Late Quaternary Records of Environmental Change from the Central Queensland Coast and Southern Great Barrier Reef
11:45	11:45–12:00		**Donald Maute**	The University of Melbourne	Preliminary insights on structurally controlled gold mineralization in the Archean Youanmi gold deposit, Yilgarn Craton, Western Australia		11:50–12:00	**Valerie Shayne Olfindo**	University of Queensland	Brisbane, QLD	Mafic magma recharge driving the Taal 1965-1969 eruption recorded in clinopyroxene crystals
12:00	12:00–12:15		**Felix Igede**	The University of Melbourne	Structural controls on sulphide mineralisation at the JIM hydrothermal field, Southeast Indian Ridge	⇄	12:00–12:10	**Shubhadeep Roy**	Queensland University of Technology	Brisbane, QLD	How hot was the Archaean mantle? Thermodynamically self-consistent benchmarks from modern basalts
12:10							12:10–12:20	**Blake Forrest**	University of New England	Armidale, NSW	Revealing the cranial anatomy of the Western Australian Triassic rhytidosteid temnospondyl Deltasaurus pustulatus (Stereospondyli, Trematosauria) via synchrotron scanning.
12:15	12:15–12:30		**Isabella Duncan**	Monash University	U-tolerant bacteria: The biogeochemical cycling of U, P, and REEs		12:20–12:30	**Henry Edward Calvert Grant**	University of New England	Armidale, NSW	Multi-proxy chronostratigraphy of the Cambrian (Miaolingian–Furongian) Arthur Creek Formation near Boulia, western Queensland
12:30	12:30–12:45		**Foojan Kazemzadeh Haghighi**	The University of Melbourne	Heterogeneity of pore geometry and directional fluid flow behavior in vesicular basalts	⇄	12:30–12:40	**Ethan Downie**	University of New England	Armidale, NSW	A diapsid braincase from the Late Cretaceous of Central Queensland
12:40							12:40–12:50	**Josh Gardiner**	University of New England	Armidale, NSW	On the Biogeographic Conundrum that is Serendipaceratops
12:45	12:45–13:00		**Rhiannon Ferdinando**	The University of Melbourne	Legacy Well Assessment in a Saline Aquifer: Beagle Sub-basin, Australia		12:50–13:00	**Ian Rogers**	University of Western Australia / CSIRO	Perth, WA	Understanding hydrogen and helium systems on the Eyre Peninsula, South Australia: what can the water tell us?
13:00	LUNCH · both rooms · break time fixed by the GESSS national stream										

ROOM 1 · VUEESC

Time	Slot	Session	Speaker	Institution	Title
13:45–14:30	VUEESC KEYNOTE · Juliet Sefton · title TBC · Room 1, all VUEESC delegates				
14:30	14:30–14:45	S2 Mantle, magmas and meteorites	**Charles Lacy**	The University of Melbourne	Petrogenesis of the diamondiferous Wolf 'kimberlites' in Finland
14:45	14:45–15:00		**Ying Ying Choi**	The University of Melbourne	Genesis of Orbicular Kimberlites: Implications for Magma Composition and Evolution
15:00	15:00–15:15		**Rohan Li-Zhi Anderson**	The University of Melbourne	High-Resolution Geochronology and Geochemical Characterisation of the Youngest Tuffs of the Turkana Basin, Kenya
15:15	15:15–15:30		**Kensaku Moriya**	The University of Melbourne	Application of X-ray Micro Computed Tomography for Meteorite and Space Mission Returned Sample Analysis
15:30	15:30–15:45		**Hugo Gouilloux**	The University of Melbourne	Evaluating elemental contamination from personal protective equipment and consumables within clean labs
15:45–15:55	AFTERNOON TEA + POSTER · both rooms · VUEESC poster: Jacob Wilson				
16:00–16:15	GESSS KEYNOTE · Wolfgang Preiss · streamed in Room 1				
16:20	16:20–16:35	S3 Air quality, weather and health	**Aberta Rulinri Siahaan**	The University of Melbourne	Understanding ozone formation in Melbourne
16:35	16:35–16:50		**Wattanapon Tiempathom**	The University of Melbourne	Elevated Aerosol Layers over Melbourne: Characteristics and Potential Drivers from MAX-DOAS Observations
16:50	16:50–17:05		**John Drugli**	The University of Melbourne	The Meteorology of Thunderstorm Asthma events in Victoria, Australia
17:05	17:05–17:20		**Salman Khan**	The University of Melbourne	Rethinking Ventilation Rates Through Infection-Risk Modelling: A Room-Specific Framework for Healthier Buildings
17:20	17:20–17:35		**Isaac Benjamin Miles-Fine**	The University of Melbourne	The first generalised ensemble tangent linear model for automatic model differentiation.
17:35	VUEESC Day 1 finishes · Trivia night at the Clyde Hotel from 5:30 pm (sponsored by GSAV)				
17:40	*End of day*				

ROOM 2 · GESSS Nationwide (live stream)

Slot	Speaker	University	Presenting from	Title
13:45–13:55	**Matthew Zhang (Xiaodong)**	University of Western Australia	Perth, WA	A multi-billion year crustal record from the northwestern margin of the West African Craton
13:55–14:05	**Heidi Jane Allen**	University of Melbourne	Perth, WA	From Alcheringa to beyond
14:05–14:15	**Rhys Meyerkort**	Curtin University	Perth, WA	Upper Cretaceous pliosaurid teeth from Western Australia
14:15–14:25	**Andrew Ji Yao Jian**	Curtin University	Perth, WA	Microbial sulfur redox cycling controls barite–calcite–pyrite permineralisation of three-dimensionally preserved ichthyosaur bones in a carbonate concretion
14:25–14:35	**Morgane L. Perret Lévesque**	Curtin University	Perth, WA	Geological, palaeoenvironmental and palaeontological constraints on the fossil record of Meiolania platyceps on Lord Howe Island (Pleistocene, Pacific Ocean)
14:35–14:45	**Helen Ryan**	University of New England	Perth, WA	Quantifying shape of Albian dimitobelid belemnites from the Gearle Siltstone, Western Australia, reveals microevolution across two species.
14:45–14:55	**Ashlea Turner**	James Cook University	Townsville, QLD	The Largest Stereospondyl (so far) from the Late Triassic Arcadia Formation (Bowen Basin) of Southeast Queensland
14:55–15:05	**Tabitha Colborne**	James Cook University	Townsville, QLD	What drives the transition from wide to narrow continental rifting?
15:05–15:15	**Elizabeth Marino**	James Cook University	Townsville, QLD	Optical microscopy for identifying deformation of sulphides in ore deposit studies
	GESSS afternoon break (from 15:15)			
16:00–16:20	**Wolfgang Preiss**	Keynote	Adelaide, SA	**KEYNOTE: Toward a Four-Dimensional Conceptual Model of the South Australian Cryogenian to Ediacaran Heysen Supergroup**
16:20–16:30	**Xavier James Damiani**	Adelaide University	Adelaide, SA	Palaeobiology of Phyllozoon hanseni from the Ediacaran of South Australia
16:30–16:40	**Joshua Batt**	Flinders University	Adelaide, SA	The rhizodontids and tetrapods (Tetrapodomorpha, Sarcopterygii) from the Drummond Basin, Queensland: International significance for stem tetrapod evolution.
16:40–16:50	**Thomas Turner**	Flinders University	Adelaide, SA	Exploring the taphonomy of the exceptionally preserved arthropod fossils from the Gogo Formation
16:50–17:00	**Darius Boast**	Adelaide University	Adelaide, SA	The Skillogalee Dolomite in a changing Neoproterozoic world: a case study from the northern Mt Lofty Ranges
17:00–17:10	**Mariella Greenaway**	Adelaide University	Adelaide, SA	Dating multi-generation hydrothermal/brine fluid flows via in-situ Rb-Sr in the Mt Lyndhurst Diapir, Northern Flinders Ranges, South Australia.
17:10–17:20	**Gabriel Cellier**	James Cook University	Townsville, QLD	Sulfides as tracers of critical metals and ore evolution: a case study from the Mount Colin ISCG deposit, Cloncurry Province.
17:20–17:30	**Sebastian Jurado Hernandez**	James Cook University	Townsville, QLD	Refining Tephrostratigraphic Correlation Methods Using Zircon Geochemistry and Hosted Melt Inclusions Compositions from Taupō Volcanic Zone deposits.
17:30–17:40	**Nicko Wyndham**			Closing remarks

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GESSS Melbourne hub, in person: the only room running

	ROOM 1 · VUEESC						ROOM 2 · GESSS Nationwide (live stream)				
Time	Slot	Session	Speaker	Institution	Title	⇄	Slot	Speaker	University	Presenting from	Title
9:00	Welcome back and morning coffee · Room 1, all delegates										
9:30	9:30–9:45	S4 Coasts, lakes and people	**Rita Fortune**	The University of Melbourne	Interpretations and implications of the textures and mineralogy of microbialites in Western Victoria		*GESSS stream starts at 11:00*				
9:45	9:45–10:00		**Ahmed Naceur MAMA**	The University of Melbourne	A modeling perspective on hydro-climate variability in dryland lakes: Controls on the rise, persistence and demise of Kati Thanda-Lake Eyre						
10:00	10:00–10:15		**Thomas Faraon**	The University of Melbourne	Process-based modelling of wind waves in a data-scarce lake: Insights from Kati Thanda–Lake Eyre, Australia						
10:15	10:15–10:30		**Sophie Golding Chan**	The University of Melbourne	Timing and magnitude of Holocene relative sea-level and coastal changes on the southern coast of Australia.						
10:30	10:30–10:45		**Tehreem Raza**	The University of Melbourne	Reconstructing Holocene Environments and Seasonal Foraging Practices through Mollusc Shell Sclerochronology in Western Victoria, Australia						
10:45	VUEESC TEA BREAK										
11:00							11:00–11:10	**Nicko Wyndham**			Opening remarks
11:15	11:15–11:30	S5 Quaternary archives and chronology	**Ngozi Ulasi**	The University of Melbourne	Radiometric age estimates for Terminations XVI and XVII: implications for orbital forcing.		11:10–11:20	**Sailakshmi Cholanilath**	University of New South Wales	Sydney, NSW	Early earth Microbialites and clues to evolution of complex life
11:30	11:30–11:45		**Louisa Sheridan**	The University of Melbourne	Investigating the impact of millennial scale climate events on southern Australia during the Last Glacial Period.	⇄	11:20–11:30	**Christopher D. Vournazos**	University of Wollongong	Sydney, NSW	Reconstructing dinosaur diets in ancient ecosystems using multi-proxy tools
11:45	11:45–12:00		**Ajisaka Octawiyano**	The University of Melbourne	Tracing Quaternary Fluvial Evolution through Remote Sensing, Sedimentology, Geochemistry, and Luminescence Geochronology in the Werribee River		11:30–11:40	**Rafael Cherene**	University of Sydney	Sydney, NSW	Linking Stratigraphic Forward Modelling to Hydrogen Mobility: Implications for Underground Hydrogen Storage in a Data-Limited Passive Margin
12:00	12:00–12:15		**Harriet Magee**	The University of Melbourne	Fire Intensity Sensitivity of Stable Carbon Isotope Composition and Chemical Signature of Charcoal	⇄	11:40–11:50	**Chamini Sandunika Waravita**	University of Wollongong	Sydney, NSW	Investigating Mine-Based Geothermal Energy Potential in the Sydney Basin, Australia: A Geospatial Framework.
12:15	12:15–12:30		**Mfoniso Anieti Otu**	The University of Melbourne	Two Antarctic Cores, One Magnetic Story?		11:50–12:00	**Puqing Li**	University of Sydney	Sydney, NSW	From rocks to geophysics: petrophysical properties of mantle xenoliths from Mount Wiridgil, southeast Australia
12:30	12:30–12:45	S6 Weather extremes and water (part 1: extremes and hazards)	**Letitia Solly**	The University of Melbourne	Increasing Summertime Afternoon Hourly Rainfall Extremes in Temperate Eastern Australia	⇄	12:00–12:10	**Caesar Augusto Rigoti**	University of Sydney	Sydney, NSW	Rift–Plume Interaction and Microplate Evolution in the South Atlantic: The Rio Grande Rise–Valdivia Bank System
12:45	12:45–13:00		**Si Phong (Rowland) Nguyen**	Monash University	Beyond Single-View Wildfire Susceptibility Mapping: A Trust- and Uncertainty-Aware Multi-View Framework		12:10–12:20	**Penelope Beltran-Rehberg**	University of Sydney	Sydney, NSW	Investigating Granulite Facies Metamorphism in Paleoproterozoic Rifted Basins, with examples from Broken Hill, New South Wales
							12:20–12:30	**Rory Carter**	University of New South Wales	Sydney, NSW	Deciphering cryptic rare earth element enrichment in the skarn regolith at Doradilla.
12:30						⇄	12:30–12:40	**Michael Fakes**	Adelaide University	Adelaide, SA	The Garnet Factory: A HREE Reservoir
12:45							12:40–12:50	**Holly Cooke**	Adelaide University	Adelaide, SA	Enigmatic origins of the rarest, most economic magmas on the planet: the aillikite-carbonatite connection in central Australia's newly discovered Nb-REE alkaline province
13:00							12:50–13:00	**David Baker**	Adelaide University	Adelaide, SA	Carbonatite melt pathways connect the deep mantle to Archean craton margins.
13:15	LUNCH · both rooms · break time fixed by the GESSS national stream										
13:30											

Time	ROOM 1 · VUEESC					⇄	ROOM 2 · GESSS Nationwide (live stream)				
	Slot	Session	Speaker	Institution	Title		Slot	Speaker	University	Presenting from	Title
13:45						⇄	13:45–13:55	**Evandro Augusto de Souza Magalhaes**	Adelaide University	Adelaide, SA	Reconstructing Holocene fire history on Kangaroo Island: a biomarker-led multiproxy approach.
13:50	*Room 1 free · lunch continues until the Melbourne hub talks*										
13:55							13:55–14:05	**Teye Frederik Nicolaas Aukes**	Adelaide University	Adelaide, SA	Inferring climate change in south-west Western Australia using diatoms from Boat Harbour Lagoon and the subsequent ecological response through sedimentary ancient DNA
14:00											
14:05	14:05–14:15	**GESSS Melbourne hub** **In person, streamed nationally**	**Boying Lyu**	University of Melbourne	Drought-to-wet whiplash in Australia: historical changes across timescales		*Room 2 not in use · the GESSS Melbourne hub talks are in Room 1 and streamed nationally from there*				
14:10											
14:15	14:15–14:25		**Natasha Nagle**	La Trobe University	Shifting Shorelines: An Exploration of Late Pleistocene Lacustrine Deposits in the Jordan Valley and their Environments						
14:20											
14:25	14:25–14:35		**Yangxue Chen**	University of Melbourne	Spatial and temporal variability in wave storm characteristics across Port Phillip Bay, Australia						
14:30											
14:35	14:35–14:45		**Tom McNamara**	University of Melbourne	Uses and interpretation of muographic opacity data as a geologist						
14:40											
14:45	14:45–14:55		**Ithay Rott**	Monash University	Redefining the source of orogenic gold deposits: The role of deformation in driving the pyrite-pyrrhotite transition						
14:50											
14:55	14:55–15:05		**Ishara Pathirage**	University of Melbourne	From gabbro weathering to rock varnish: Mineral-scale element redistribution at Murujuga, Western Australia						
15:00											
15:05	15:05–15:20	**S6** **Weather extremes and water (part 2: water and drought)**	**Afifudin**	The University of Melbourne	Spatial Distribution of Water Demand for Domestic Use in a Disaster-Prone Coastal City: A Case Study of Pekalongan City, Indonesia	⇄	15:05–15:15	**Eleanor V. Y. Beidatsch**	University of New England	Albany, WA	Walking “Worms” Part: Describing the First Occurrences of the Cambrian Armoured Lobopodian Quadratapora from South Australia and Kazakhstan
15:10											
15:15											
15:20	15:20–15:35		**Ruchit Sandip Kulkarni**	The University of Melbourne	Atmospheric processes and meteorological droughts: from development to recovery		*GESSS afternoon break (from 15:15)*				
15:25											
15:30											
15:35	**AFTERNOON TEA · both rooms**										
15:40											
15:45											
15:50											
15:55											
16:00	**GESSS KEYNOTE · Jodie Fox · streamed in Room 1**					⇄	16:00–16:20	**Jodie Fox**	Keynote	Hobart, TAS	**KEYNOTE: Volcanoes, Ice, and an Unconventional Career Path: Marine and Antarctic Geoscience**
16:05											
16:10											
16:15											
16:20	16:20–16:35	**S7** **Climate modes and the cryosphere**	**Flynn Thompson**	The University of Melbourne	El Niño-Southern Oscillation Dynamical Response to Volcanic Forcing during the Last Millennium	⇄	16:20–16:30	**Will Sinclair**	University of Tasmania	Hobart, TAS	Eocene Coral Sea Tectonics - Constraints from the Mellish Rise, Louisiade Plateau, and Louisiade Trough
16:25											
16:30							16:30–16:40	**Maelys Moulin**	University of Tasmania	Hobart, TAS	Beyond rounding: Pumice Shape Evolution and Ash Production in Subaqueous Environment
16:35	16:35–16:50		**Jack Travers**	Monash University	Covariability in large-scale climate modes and implications for Antarctic surface mass balance variability						
16:40							16:40–16:50	**Kaitlyn Ainscow**	University of Tasmania	Hobart, TAS	Geochronological Constraints for Tasmanian Stratigraphy and Snowball Earth Events
16:45											
16:50	16:50–17:05		**Tirthankar Ghosh**	Monash University	Controls on Himalayan glacier thermal regimes revealed through ensemble thermomechanical modelling	⇄	16:50–17:00	**Gabrielle Kaye O'Toole**	University of Tasmania	Hobart, TAS	A Not So Ordinary Chondrite
16:55											
17:00							17:00–17:10	**Turki Moaaf**	ANU	Canberra, ACT	Parameterizing Lunar Seismic Scattering
17:05	17:05–17:20		**Fakhra Muneeb**	The University of Melbourne	From Mine to Melt: Global-Scale Evidence Linking Mining Proximity to Glacier Albedo Decline						
17:10							17:10–17:20	**Katya Orlova**	ANU	Canberra, ACT	An appetite for apatite: Investigating apatite geochemistry as an exploration tool for carbonatite rare earth element and niobium deposits
17:15											
17:20	**ROOM 1 PAUSED · GESSS closing & national prizes · stream in Room 1**					⇄	17:20–17:30	**Nicko Wyndham**			Closing remarks and announcing prize winners
17:25											
17:30	**VUEESC closing & prizes · Room 1, all delegates**										
17:35											
17:40											
17:45											
17:50											
17:55											
18:00	*End of day*										